

On the Signed Domination Number of Torus Graphs

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Abstract

Let $G = (V, E)$ be a simple graph. The signed dominating function assigns 1 and -1 to each vertex such that the sum of the assigned values of the closed neighborhood $N[v]$ for all $v \in V$ is more than 0. The signed domination number $\gamma_s(G)$ is the minimum sum among all possible signed dominating functions. In this paper, we give the lower and upper bounds of the signed domination number for a torus graph.

1 Introduction

Let $G = (V, E)$ be a simple graph where V is the vertex set and E is the edge set of G . As usual, let $|V| = n$. The open neighborhood of a vertex $v \in V$ is denoted by $N(v)$. In addition, $N[v] = N(v) \cup \{v\}$ denotes the closed neighborhood of v . The degree of a vertex v is denoted by $d(v)$. A set $D \subseteq V$ is a dominating set if $N[v] \cap D \neq \emptyset$ for all $v \in V$. A signed dominating is to assign each vertex 1 or -1 such that the number of the vertices assigned 1 in $N[v]$ is more than the number of the vertices assigned -1 for each vertex $v \in V$. The signed domination problem is to find a function $f : V \rightarrow \{-1, 1\}$ such that $\sum_{v \in V} f(v)$ is smallest among all possible functions. The minimum sum is called the signed domination number of G and is denoted by $\gamma_s(G)$ [3]. For simplicity, we let $f(W) = \sum_{v \in W} f(v)$. This problem also has been studied by several authors including [2, 4, 5, 6, 8, 9, 10, 12].

A regular grid graph consists of n vertices which have the same degree. If the dimension of the graph is one, this graph is a cycle C_n and is also called a ring. A torus, which is also called the toroidal, is an n -dimension regular grid graph connected circularly in more than one dimension. Tori are widely used for massively parallel computers. Such kinds of graphs have a small bisection bandwidth and can fully exploit the networks band-

width. A torus cannot only be applied in computer networks but also in parallel computing clusters and multi-core processors. In topology, a ring torus is homeomorphic to the cartesian product of two circles, $C_1 \times C_n$. A special case of torus sitting in R^4 is the Clifford torus in geometric topology. Each vertex of the Clifford torus has the same distance from the origin. Toroidal objects include doughnuts, vadais, inner tubes, bagels, lifebuoys, O-rings and vortex rings [1, 8, 11].

In this paper, we first give a general lower bound of the signed domination number of a k -regular graph. Since a torus graph, $C_r \times C_s$ where r and s are two positive integers, is 4-regular, the lower bound of the signed domination number of a torus graph can be obtained by the formula of the signed domination number of a k -regular graph. Then the upper bound of the signed domination number of a torus graph is obtained by using a signed dominating function.

2 Lower Bound and Upper Bound

In this section, we first consider the k -regular graphs. A graph $G = (V, E)$ is k -regular if $d(v) = k$ for all $v \in V$. Figure 1 gives an example.

Theorem 1. A lower bound of the signed domination number of a k -regular graph is $\frac{n}{k+1}$.

Proof. Let $G = (V, E)$ be a k -regular graph with $V = \{v_1, v_2, \dots, v_n\}$. By the definition of the signed domination, we have

$$f(N[v_i]) \geq 1, 1 \leq i \leq n. \quad (1)$$

Then we summarize Formula (1) for $1 \leq i \leq n$ and obtain

$$\sum_{1 \leq i \leq n} f(N[v_i]) \geq n. \quad (2)$$

Since G is k -regular, the degree $d(v)$ for each $v \in V$ is equal to k . Thus, for each vertex $v \in V$, it

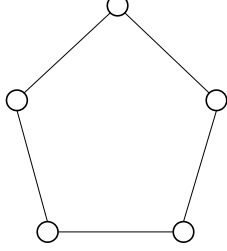


Figure 1: A 2-regular graph with five vertices.

contributes $k + 1$ times in Formula (2) since v is in the closed neighborhood of k vertices and itself. Therefore, we can rewrite Formula (2) as

$$(k + 1)f(V) \geq n.$$

Hence we have the following equation and complete our proof.

$$f(V) \geq \frac{n}{k + 1}.$$

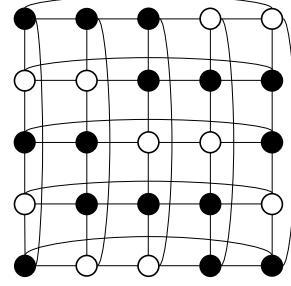
□

Since a torus graph is 4-regular, we can easily obtain the following corollary.

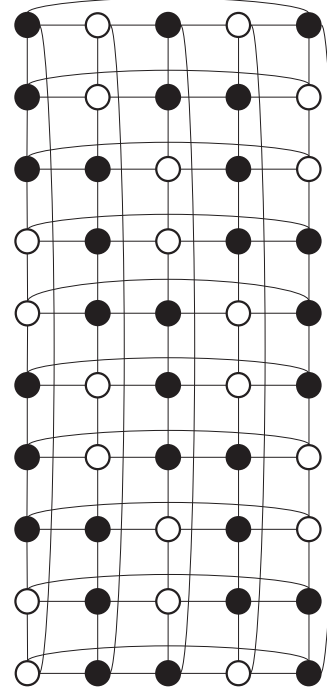
Theorem 2. *The lower bound of the signed domination number of a torus graph is $\lceil \frac{n}{5} \rceil$.*

For the upper bound, we propose an algorithm for labeling a torus graph according to rules of signed domination. Suppose that the torus graph is $C_r \times C_s$. We can assign labels to vertices row by row or column by column. Without loss of generality, we assign labels row by row. For the first row, we repeat the pattern “1, 1, 1, -1, -1.” For the second row, we shift the pattern rightward for two positions. In general, any three 1’s would follow two -1’s. Fill the leading two positions according to this rule. Similarly, for the remaining rows, labels in each row are a two positions of rightward shift. Figure 2 shows an example for $C_5 \times C_5$. In the figure, black (respectively, white) vertices denote vertices of 1 (respectively, -1).

Consider the torus graphs $C_r \times C_s$ with $r \bmod 5 = s \bmod 5 = 0$, i.e., $C_{5p} \times C_{5q}$ where p and q are positive integers. It is not hard to see that $C_{5p} \times C_{5q}$ is an extension of $C_5 \times C_5$. That is, we may see a $C_5 \times C_5$ as a unit subgraph. Thus,


 Figure 2: Our labeling method for $C_5 \times C_5$.

$C_{5p} \times C_{5q}$ contains pq $C_5 \times C_5$ subgraphs. Note that we can use the same labeling method for each subgraph. Figure 3 shows the result for our algorithm to label $C_{10} \times C_5$.


 Figure 3: Our labeling method for $C_{10} \times C_5$.

It is not hard to check that our algorithm for $C_{5p} \times C_{5q}$ is optimal and we have the following theorem.

Theorem 3. *The signed domination number of $C_{5p} \times C_{5q}$, where p and q are positive integers, is $5pq$, i.e., $\frac{n}{5}$.*

Now, we consider the case that only one of r

and s is a multiplier of 5 in $C_r \times C_s$. Without loss of generality, we assume the torus graph is $C_{5p} \times C_{5q+k}$, where $1 \leq k \leq 4$. By doing our algorithm in this case, only the last column would not obey the rules of signed domination. Since vertices in each row form a cycle, only the vertices of label -1 in the last column will be a problem. However, if we let the label of every vertex in the last column be 1, then this new labeling is a signed domination. Since r is a multiplier of 5, each column has $\frac{2r}{5}$ vertices of label -1. Thus the total value obtained by our algorithm in this case is

$$\frac{r(s-1)}{5} + r = \frac{n}{5} + \frac{4r}{5}.$$

Therefore, we have the following theorem.

Theorem 4. *The upper bound of signed domination number of $C_r \times C_s$, where r is a multiplier of 5 and s is not, is $\frac{n}{5} + \frac{4r}{5}$.*

Finally, we consider the case that both r and s are not a multiplier of 5. Similarly, for our algorithm, only the vertices of label -1 in the last column and the last row will void the rules. Thus, we can let every vertex of label -1 in the last column and row become 1. It is easy to check that the result is a signed dominating function. Of course, it is an upper bound for this case. This upper bound can be evaluated as follows.

$$\begin{aligned} f(V) &= \left\lceil \frac{(r-1)(s-1)}{5} \right\rceil + r + s - 1 \\ &= \left\lceil \frac{rs-r-s+1}{5} \right\rceil + r + s - 1 \\ &\leq \left\lceil \frac{rs}{5} \right\rceil - \left\lfloor \frac{r+s-1}{5} \right\rfloor + r + s - 1 \\ &\leq \left\lfloor \frac{n}{5} \right\rfloor + \left\lceil \frac{4(r+s-1)}{5} \right\rceil \end{aligned}$$

Therefore, we have the following theorem.

Theorem 5. *The upper bound of signed domination number of $C_r \times C_s$, where both r and s are not a multiplier of 5, is $\left\lfloor \frac{n}{5} \right\rfloor + \left\lceil \frac{4(r+s-1)}{5} \right\rceil$.*

3 Conclusion

In this paper, we propose an algorithm for the signed domination problem on torus graphs $C_r \times C_s$. In the case that both of r and s are a multiplier of 5, our algorithm correctly compute $\gamma_s(C_r \times C_s)$, namely, $\frac{n}{5}$. In the case that r is a multiplier of 5 but s is not, we obtain that $\left\lfloor \frac{n}{5} \right\rfloor \leq \gamma_s(C_r \times C_s) \leq \frac{n}{5} + \frac{4r}{5}$. If both of r and s are not a multiplier of 5, then our algorithm obtain that $\left\lfloor \frac{n}{5} \right\rfloor \leq \gamma_s(C_r \times C_s) \leq \left\lfloor \frac{n}{5} \right\rfloor + \left\lceil \frac{4(r+s-1)}{5} \right\rceil$.

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